

Customer decisions. Managerial AI adoption.

ABC Limited

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Project outcome

A tested Streamlit decision-support application combines churn scoring, viewing-hours estimation, transparent model evidence and a managerial feedback study. Both required regression models have been built and evaluated.

Model	Held-out result	Managerial interpretation
Logistic regression	Accuracy 88.5%; ROC-AUC 0.967	Useful for a synthetic-data prioritisation demonstration.
Linear regression	RMSE 12.392 hours; R ² -0.001	Does not beat the simple-average baseline; do not use for decisions.

Business question

Which customers should a manager review for possible retention outreach, what evidence supports that priority, and under what conditions would the manager accept or challenge AI advice?

Submission status

Complete: dataset audit, two models, held-out evaluation, app code, individual and batch inference, explanations, questionnaire, scenarios and response-analysis workflow.

Pending: public app publishing, genuine managerial responses, qualitative coding and final study findings. No live URL, participants or adoption results are claimed.

Dataset qualification

The matching Kaggle publisher describes synthetic data. This is disclosed throughout. If the instructor interprets the real-world requirement as genuine observed records, dataset acceptance or replacement is still required. The original data is retained as requested.

Know what the data can support

The supplied CSV contains 5,000 customer records and 14 fields. There are 2,515 churn labels (50.3%) and 2,485 retained labels. The audit found no null cells, duplicate rows or duplicate customer IDs. The case is presented as ABC Limited.

Field(s)	Treatment	Reason
customer_id	Retain only for audit/splits	Identifier, not a meaningful predictor.
watch_hours	Churn input; linear target	Recorded viewing volume; observation window not specified.
last_login_days; number_of_profiles	Inputs in both models	Available engagement and account attributes.
subscription_type; region; device; payment_method; favorite_genre	Categorical inputs in both models	One-hot encoded using training data only.
monthly_fee	Exclude from modelling	Exactly fixed by plan in this file; duplicates plan information. Currency is not documented.
avg_watch_time_per_day	Exclude from both models	Matches watch_hours/(last_login_days+1) within rounding for all rows; 10 values exceed 24.
age; gender	Retain in source, exclude from prediction	Not needed for this operational prototype. This does not by itself prove fairness.
churned	Logistic target only	Never used as an input to the linear model.

Scope limitations

There are no observation timestamps, tenure measures, billing histories, customer-support records, actual retention outcomes or defined prediction horizons. The classifier predicts the recorded label; it is not validated as a next-month churn forecast. The linear model estimates recorded hours in an unspecified period, not future demand.

Source attribution

User-supplied CSV, matched to a Kaggle listing by Abdul Wadood that describes synthetic customer behaviour. The public listing matches the name, schema and size, but an independent byte-for-byte source comparison was not performed. Full source link and file hash are in SOURCE.md.

Separate learning from evaluation

Customers are randomly split, stratified by churn, into 3,000 training, 1,000 validation and 1,000 test records using seed 42. This is appropriate for a snapshot demonstration; temporal generalisation remains untested. Each customer appears in exactly one split.

Training-only preprocessing

Numerical inputs are standardised using training means and scales. Categorical inputs are one-hot encoded with one reference category dropped. Preprocessing is inside each model pipeline. Raw records are preserved; no fabricated observations or target values are introduced.

Logistic regression

Three regularisation settings ($C = 0.1, 1$ and 10) are compared using validation ROC-AUC. $C = 10$ is selected. A threshold of 0.61 maximises validation F1 on the predefined 0.10 - 0.90 grid. The test set is used only after these choices are fixed. The saved app model is the same evaluated training-fitted model.

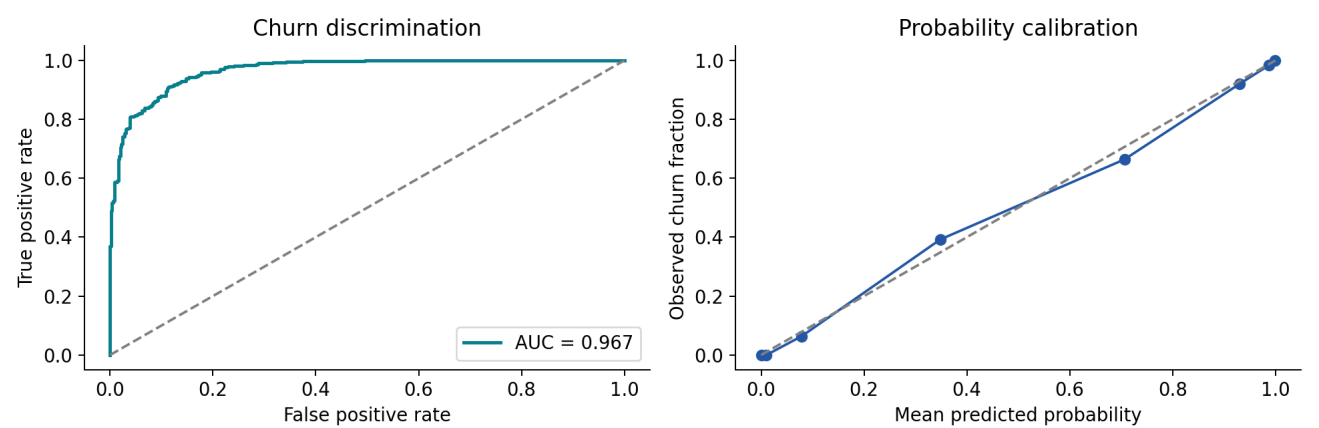
Test measure	Logistic model	Constant baseline
Accuracy	88.5%	50.3%
Precision	91.3%	50.3%
Recall	85.3%	100.0%
F1	0.882	0.669
ROC-AUC	0.967	0.500
Brier score (lower is better)	0.0729	0.2500

The confusion matrix contains 429 correctly flagged churn cases, 456 correctly retained cases, 41 false alarms and 74 missed churn cases. This error trade-off needs a business-cost assessment before operational use.

Threshold transparency

At the conventional 0.50 threshold, test accuracy is 89.3% and F1 is 0.894 . The deployed threshold remains 0.61 because it was selected on validation, not retuned after seeing test results. Maximising F1 is not the same as maximising profit.

Strong churn ranking; weak hour estimates

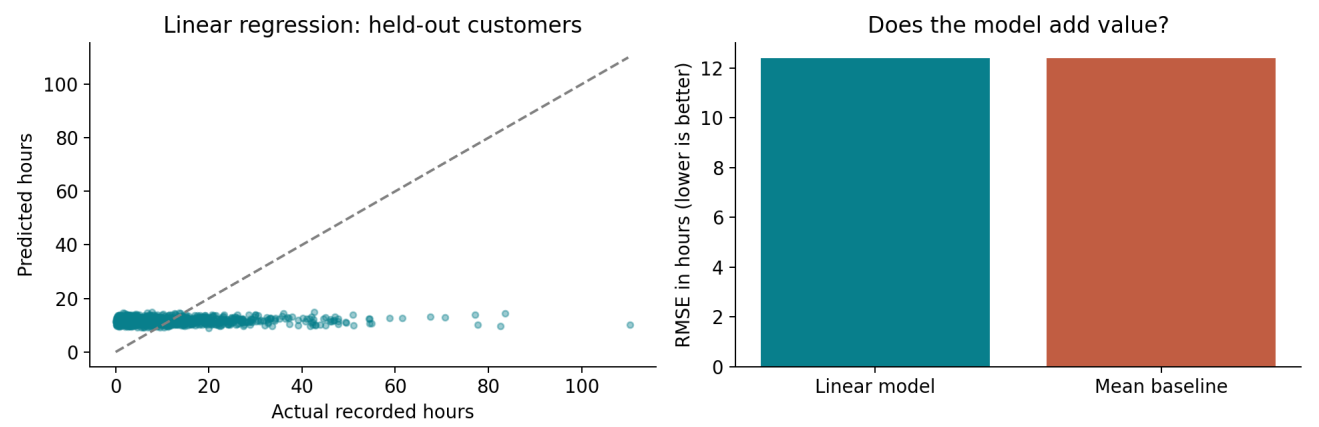


ROC-AUC measures ranking ability; it is not an accuracy percentage. The calibration plot compares probability estimates with observed churn rates in held-out groups. Both are evidence about the synthetic file, not an assurance of real-company performance.

Ordinary linear regression

The model predicts watch_hours from login recency, number of profiles and the five categorical inputs. Viewing hours, the derived daily measure, churn label, fee, identity and demographic fields are excluded from its inputs.

Test measure	Linear regression	Training-mean baseline
MAE (hours)	8.833	8.829
RMSE (hours)	12.392	12.386
R ²	-0.0010	-0.0001



The available inputs do not demonstrate useful linear predictive signal for viewing hours. The app visibly marks this estimate as unsuitable for actions. A validation-residual 90% interval has a half-width of 14.87 hours and test coverage of 90.3%; this wide interval reinforces the limitation. It relies on exchangeability and does not guarantee real-world coverage.

A manager can inspect the advice

App page	What the user does	What the tool returns
Overview	Read the business purpose and limitations	Dataset summary and the two model conclusions.
Customer check	Enter behaviour and subscription fields	Churn score, review priority, explanation and qualified viewing-hours estimate.
Batch review	Upload the supplied CSV format	Ranked customer list with downloadable scores.
Model evidence	Review evaluation before adoption	Baselines, confusion matrix, diagnostics and performance boundaries.
Manager study	Make initial judgments, see evidence and respond	A downloadable anonymous response file.
Study analysis	Upload actual response files and human coding	Rating summaries, decision changes and theme counts.

Explainability

For each churn prediction, the app decomposes the logistic score into exact feature contributions to log-odds. Positive contributions raise the score and negative contributions lower it relative to the fitted intercept and reference categories. These are associations learned by the model, not causal explanations or percentage-point effects.

Human decision process

A high score should prompt a review of recent activity and customer context. It does not justify automatic discounts, suspension or intrusive outreach. A lower score still allows churn. No retention uplift, revenue saved or financial return has been measured by this project.

What production use would require

Obtain genuine observations measured before the outcome, define a prediction horizon, add tenure and billing/support histories, evaluate on later periods, examine calibration and subgroup errors, and measure intervention benefit in a controlled pilot. The viewing-hours model needs stronger independent predictors or a better dataset.

Deployment and persistence

The package runs on Python 3.12 using pinned dependencies and saved pipelines. Publishing requires the user's GitHub/Streamlit account. No live URL exists in this package. The survey deliberately uses a download-and-return workflow; responses are not centrally stored by the app.

Observe judgment before and after advice

Study purpose

Explore usefulness, trust, willingness to adopt, explainability, conflicts with experience, fear of wrong decisions and organisational barriers. The study is exploratory and qualitative, supported by descriptive ratings.

Participants

Suggested target: 8-12 experienced decision-makers, subject to the instructor’s requirements. Relevant roles include managers, professionals, team leaders, supervisors and entrepreneurs. The provided assignment does not specify a minimum. Report actual achieved participation honestly.

Stage	Standard procedure
Consent and context	Obtain voluntary consent for anonymised responses and quotes. Record role, experience and decision context without names or employers.
Initial judgment	Begin directly at Manager study. Participants classify the retention-review priority of fixed customers A, B and C before seeing scores.
Evidence review	Reveal the same model scores, explanations, performance and synthetic-data limitations. Cases come from validation examples across score levels.
Revised judgment	Ask for final Low/Medium/High priorities and an explanation of each change or unchanged decision.
Feedback	Collect six ratings and nine open-ended responses. Use neutral follow-up prompts.
Receipt	Participant downloads the final CSV and privately returns it to the researcher. Confirm receipt before counting a completed response.

Research integrity

No interviews have been conducted by the code and no responses are included. Test fixtures are explicitly marked as software tests and are never exported into the actual study. Keep original returned files privately and honour withdrawal requests using the anonymous respondent ID.

Interpretation boundaries

A change in priority does not prove a better decision. This before/after task does not isolate the effect of explanations from the score or other disclosed information. A small purposive sample cannot establish population adoption rates or causal relationships.

Ask about use, trust and disagreement

Ratings: 1 strongly disagree to 5 strongly agree

1. This tool would help me make customer-retention decisions.
2. I found the tool easy to understand and use.
3. I trust the churn score enough to consider it alongside other evidence.
4. The explanation helped me understand the churn score.
5. I would consider using a validated version of this tool at work.
6. Concern about making a wrong decision would discourage my use.

Open-ended questions

1. For which tasks would this tool be useful, and where would it not help?
2. What did you like about the tool?
3. What did you dislike or find confusing?
4. Why did you trust or distrust the prediction? What evidence would increase your trust?
5. If your experience and the model disagree, what would you do, and why?
6. Did seeing the explanation affect your willingness to use the tool? How?
7. What personal or organisational factors would discourage adoption?
8. The viewing-hours model performed worse than a simple average. How should a manager respond?
9. What would need to change before you would use this tool at work?

For each scenario, also ask: What is your priority before seeing the model? What is your final priority after reviewing it? Why did you keep or change your decision?

Neutral probes: "Can you give an example?" "What makes you say that?" "What other information would you need?" Avoid suggesting that adoption or agreement is the desired answer.

Turn real responses into findings

Descriptive analysis

The app validates consent, rating ranges and respondent IDs, removes exact duplicates and blocks conflicting duplicate responses. It produces rating counts, medians, means, agreement percentages and scenario decision-change counts. Keep the error-concern item separate because its direction differs from the favourable ratings.

Qualitative analysis

Read every response and assign one or more codes: USEFULNESS, TRUST, EXPLAINABILITY, HUMAN_JUDGMENT, ERROR_ACCOUNTABILITY, ORGANISATION, USABILITY, DATA_LIMITS or OTHER. Add an interpretation and exact supporting quote. The tool checks that quotations occur in the original response and counts distinct respondents per theme.

For each theme, report the count and sample size, one exact anonymous quote, a contrasting view where present, your interpretation and a practical implication. Multiple themes may apply to one respondent, so shares may exceed 100% in total. Do not infer themes automatically from ratings alone.

Assignment requirement	Current status
Select a suitable dataset	Selected and audited; synthetic-source qualification disclosed.
Build linear and logistic regression	Complete, with held-out evaluation and baseline comparisons.
Build a non-technical predictive tool	Complete and locally tested.
Deploy and submit app URL	Pending publishing through the user’s account.
Collect managerial feedback	Protocol and app ready; actual responses pending.
Submit survey data and analysis	Processing tools ready; actual data, coding and findings pending.

Next actions before presentation

Publish the app using DEPLOYMENT.md; verify access from an incognito window; collect actual responses; run Study analysis; complete human coding; update study/FINDINGS_TEMPLATE.md and this report with the achieved findings. If feedback is not collected in time, report it as pending.

Source references

Dataset: matched Kaggle publisher listing by Abdul Wadood; full URL and hash in SOURCE.md. All results in this report are computed from the supplied file.

[Streamlit deployment documentation](#) | [scikit-learn preprocessing and leakage guidance](#). Reviewed 27 September 2026.